

MONTE CARLO IN JULIA

An aerial photograph showing a residential neighborhood completely inundated with floodwater. The water is a murky, brownish-grey color, reaching up to the roofs of many houses and surrounding the trees. The houses have various roof colors, including grey, brown, and red. The trees are mostly green, though some appear to be partially submerged. A central road runs vertically through the middle of the image, also flooded. In the upper right corner, a green area with a baseball field is visible, partially surrounded by water.

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CEVE 543

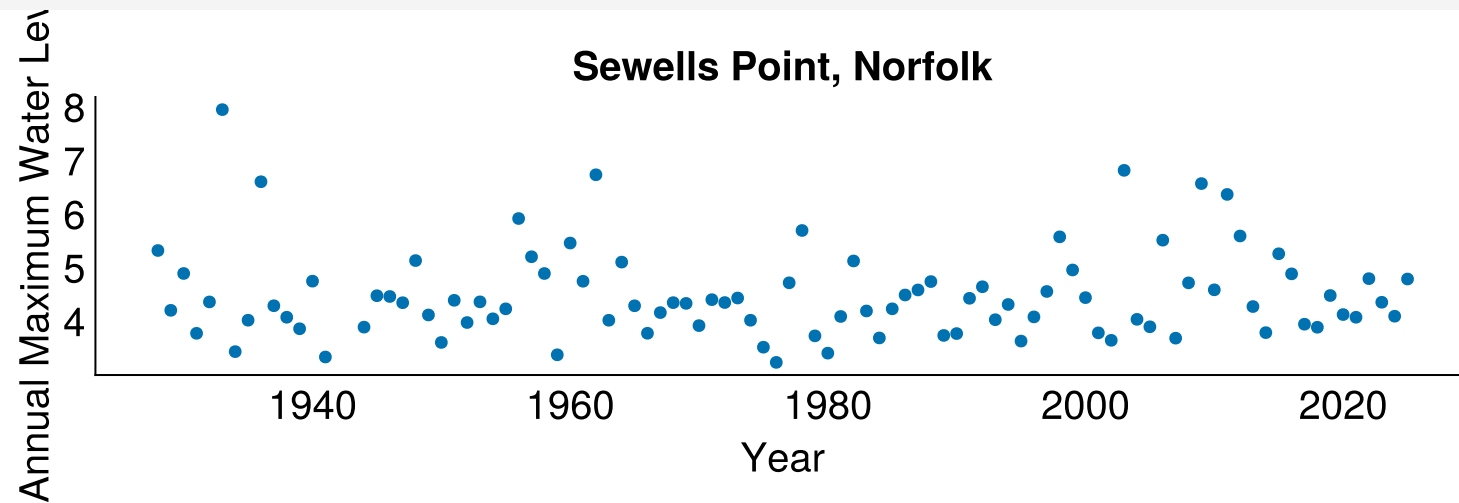
2026-08-28

Hazard

Risk

REALISTIC DATA

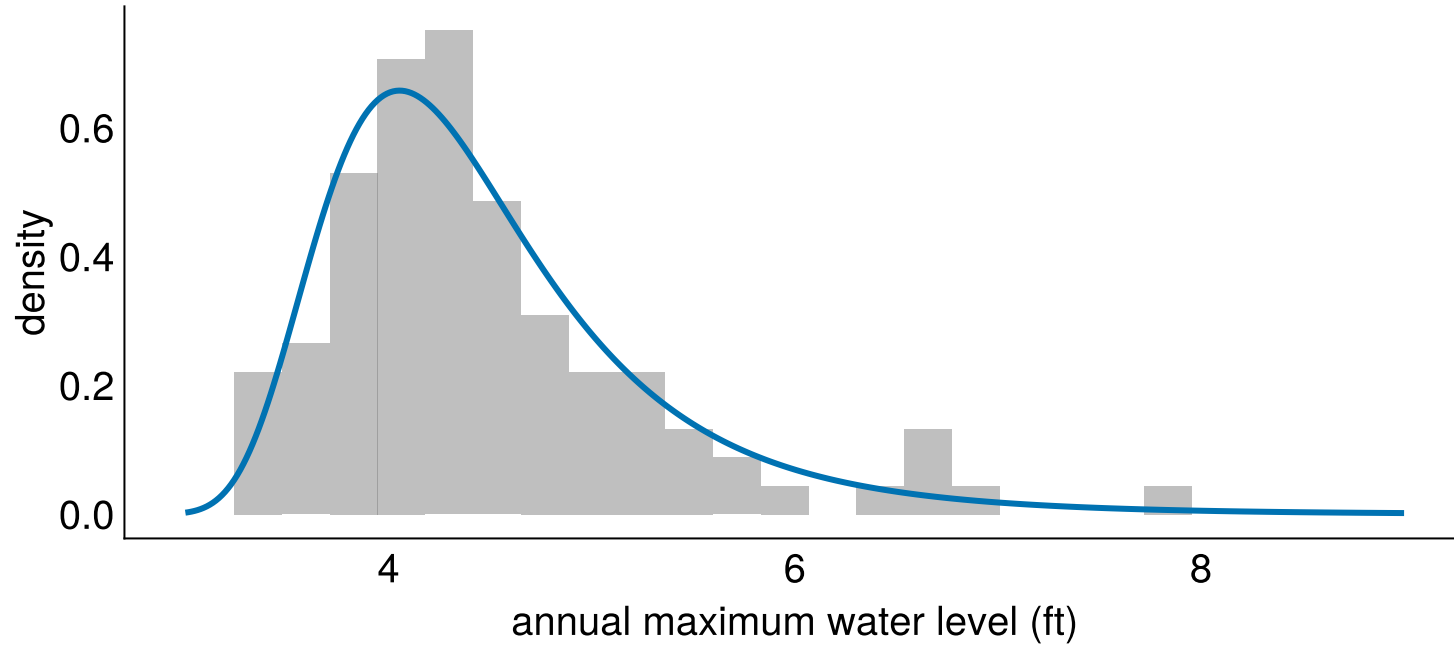
```
1 rng = MersenneTwister(824)
2 annmax = CSV.read("8638610-annmax.csv", DataFrame)
```



We fit a Generalized Extreme Value distribution (more on that soon!)

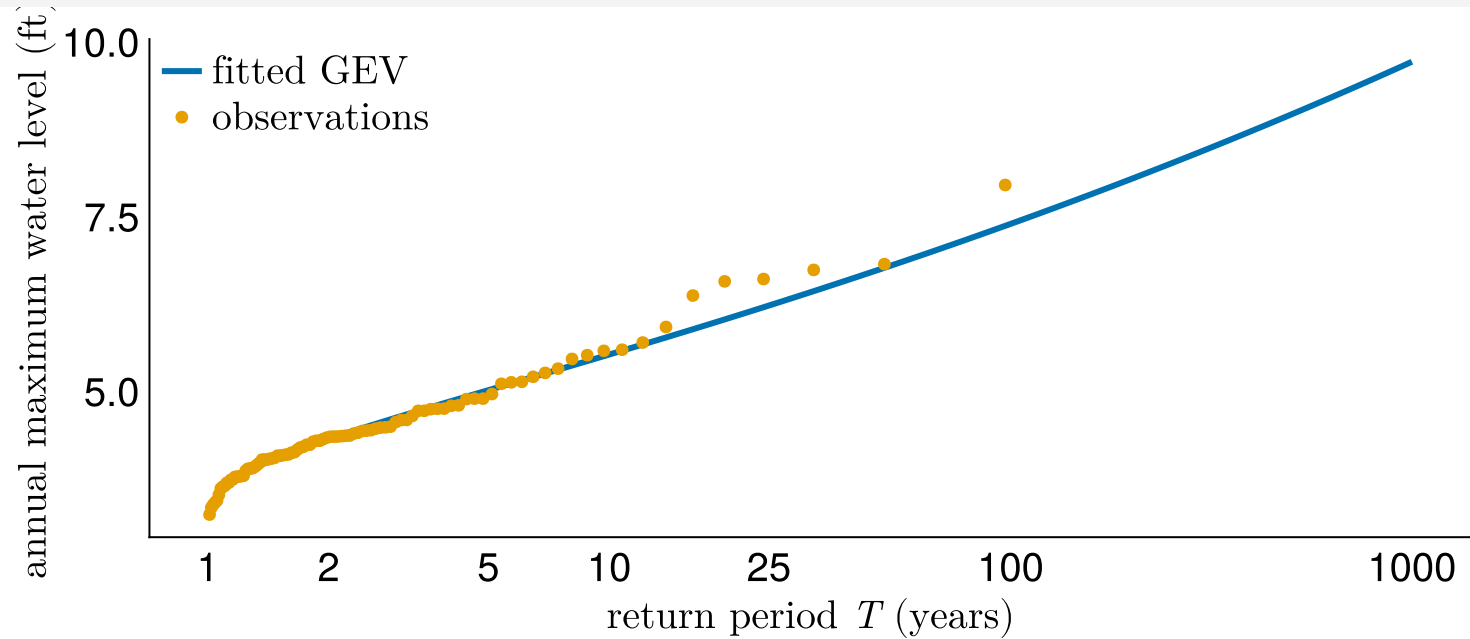
```
1 gev = only(getdistribution(gevfit(annmax.lsl)))
2 #| echo: false
3 fig = Figure(; size=(760, 350))
4 ax = Axis(fig[1, 1]; xlabel="annual maximum water level (ft)", ylabel="density")
5 hist!(ax, annmax.lsl; bins=20, normalization=:pdf, color=(:gray, 0.5))
6 lines!(ax, range(3, 9; length=400), pdf.(gev, range(3, 9; length=400)); linewidth=3)
7 fig
```

FITTING



RETURN LEVELS

```
1 n_obs = nrow(annmax)
2 lsl = sort(annmax.lsl; rev=true)
3 p_obs = (1:n_obs) ./ (n_obs + 1)
```



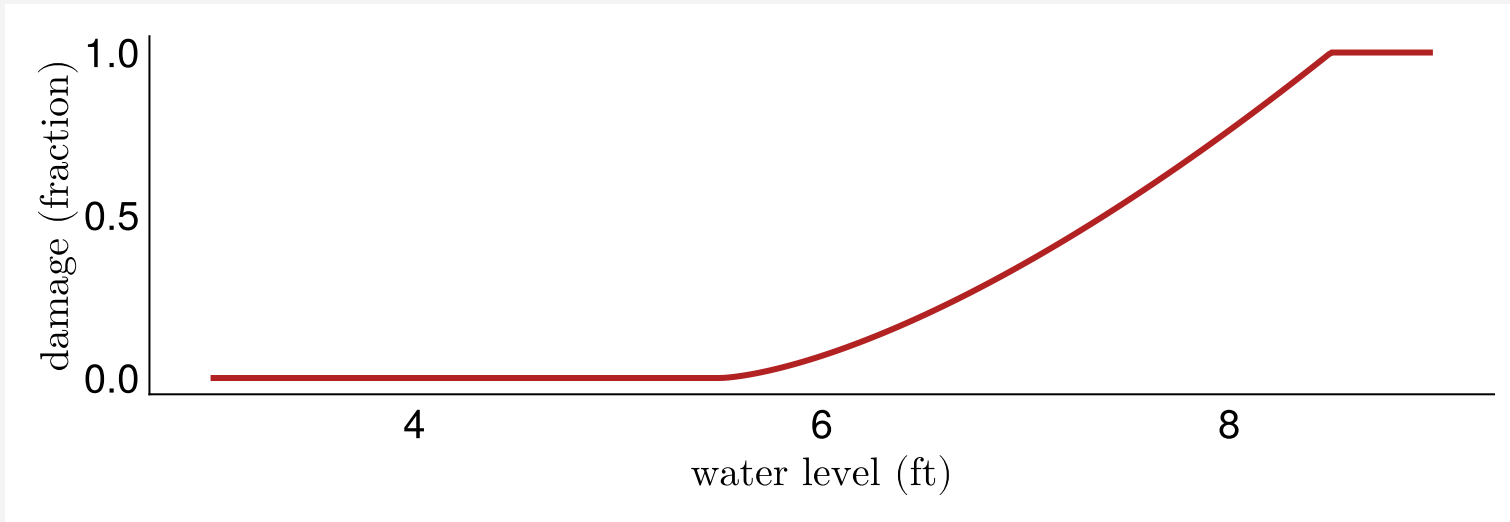
Hazard

Risk

ILLUSTRATIVE DAMAGE FUNCTION

Let h be the water level at the gauge, in feet. A building sits 5.5 ft above the gauge. Assume total loss at 8.5 ft; between those two the curve is convex.

```
1 damage(h) = clamp((h - 5.5) / 3, 0, 1)^1.5
```



EXPECTED DAMAGE WITH $n = 10$

```
1 levels = rand(rng, gev, 10)
2
3 println("level   ", round.(levels; digits=2))
4 println("damage  ", round.(damage.(levels); digits=2))
5 @printf("average %.3g", mean(damage.(levels)))
```

```
level   [4.98, 6.16, 5.69, 3.39, 5.39, 3.94, 4.56, 3.99, 5.17, 4.19]
damage  [0.0, 0.1, 0.02, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
average 0.0119
```

THE ESTIMATOR

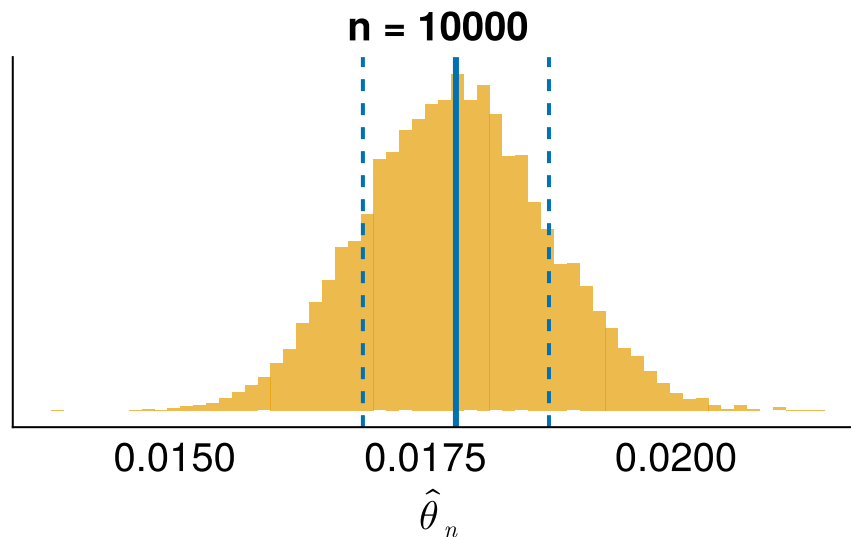
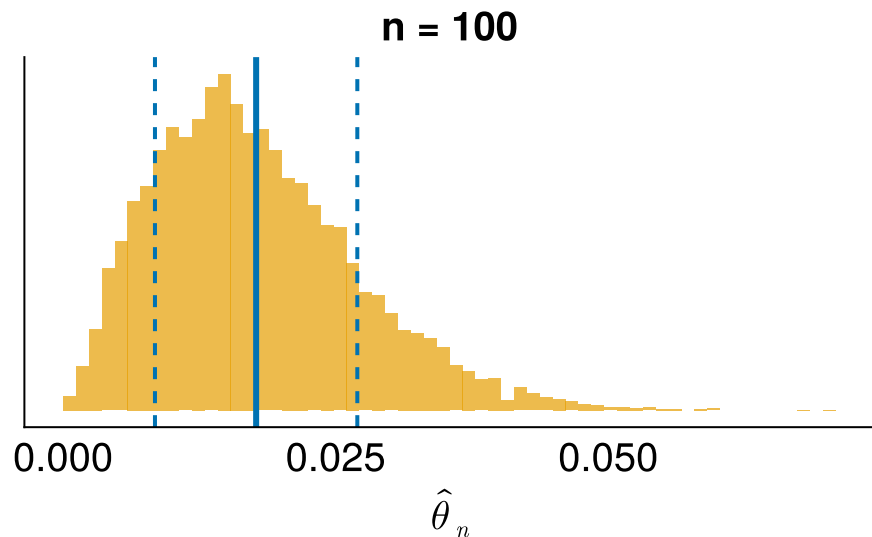
$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n g(X_i)$, with X the water level and g the damage function, estimates $\theta = E[g(X)]$:

1. Draw n years.
2. Apply g to each.
3. Average.

```
1 function expected_damage(dist, n)
2     levels = rand(rng, dist, n)
3     return mean(damage.(levels))
4 end
```

$\hat{\theta}_n$: A RANDOM VARIABLE WITH $SE = sd(g(X))/\sqrt{n}$

```
1 sd_g = std(damage.(draws))
2 sample_sizes = (100, 10_000)
3 estimates = [[expected_damage(gev, n) for _ in 1:10_000] for n in sample_sizes]
```



HOW MANY YEARS DO WE NEED TO SIMULATE?

From the whiteboard, we know we need $n = (CV/\varepsilon)^2$ simulations, where ε is one standard error expressed as a fraction of the answer.

```
1 cv(v) = std(v) / mean(v)
2 n_years(c, ε) = ceil(Int, (c / ε)^2)
```

COMPARING SIMULATIONS

How many draws do we need to estimate damage versus hazard with $\varepsilon = 0.01$?

```
1   $\varepsilon$  = 0.01
2  cv_damage = cv(damage.(draws))
3  cv_hazard = cv(draws)
4
5  @printf("damage  CV %.3g, n %d\n", cv_damage, n_years(cv_damage,  $\varepsilon$ ))
6  @printf("hazard  CV %.3g, n %d\n", cv_hazard, n_years(cv_hazard,  $\varepsilon$ ))
```

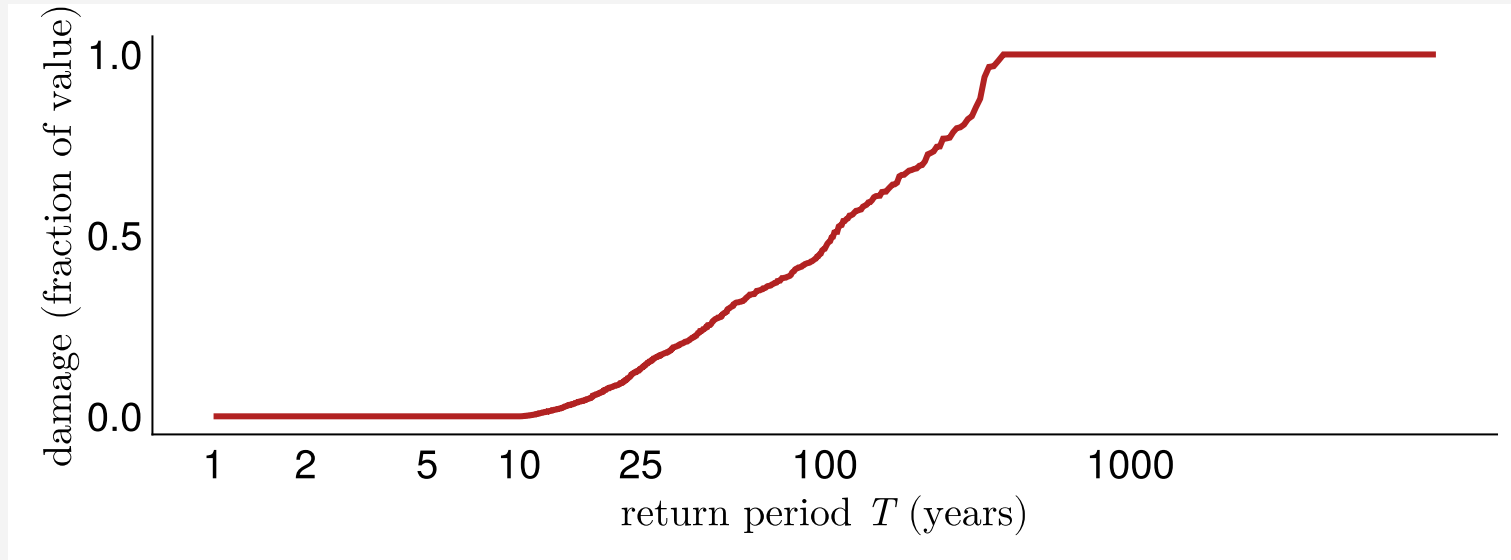
```
damage  CV 5.2, n 270669
hazard  CV 0.187, n 350
```

THE EXCEEDANCE PROBABILITY CURVE

Sort the simulated damages from largest to smallest. Rank i gets $\hat{p}_i = i/(n + 1)$ and $\hat{T}_i = 1/\hat{p}_i$ per Weibull plotting positions.

```
1 n_sim = 10_000
2 losses = sort(damage.(rand(rng, gev, n_sim))); rev=true
3 p_sim = (1:n_sim) ./ (n_sim + 1)
```


THE EXCEEDANCE PROBABILITY CURVE



IF WE KNOW F , WE CAN USE SMARTER SAMPLING

Write F for the cumulative distribution function of the water level and f for its density. Substituting $u = F(x)$ turns the integral into an average over quantiles, which we evaluate on a uniform grid of u .

$$\theta = \int d(x)f(x) dx = \int_0^1 d(F^{-1}(u)) du$$

THE TWO COMPARED AT TWO SAMPLE SIZES

```
1 function run_mc_comparison(n)
2     u = (0.5:1.0:n) ./ n
3     theta_grid = mean(damage.(quantile.(gev, u)))
4
5     @printf(
6         "n = %6d  Monte Carlo %.4g  quantiles %.4g\n",
7         n,
8         expected_damage(gev, n),
9         theta_grid
10    )
11 end
12 run_mc_comparison(1_000)
13 run_mc_comparison(10_000)
```

THE TWO COMPARED AT TWO SAMPLE SIZES

```
n = 1000 Monte Carlo 0.02049 quantiles 0.0178  
n = 10000 Monte Carlo 0.01732 quantiles 0.01781
```

WHEN MIGHT WE PREFER BRUTE FORCE MONTE CARLO?

Set your Poll Everywhere screen name to your Rice NetID so your answers are recorded.



PollEV.com/jdossgollin